

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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1 Introduction

The release of ChatGPT in November 2022 marked the beginning of large-scale diffusion of large language models (LLMs) around the world. Within months, millions of workers gained access to tools capable of drafting documents, writing code, analyzing data, and performing countless other cognitive tasks that previously required human expertise. This rapid integration into workplace workflows has generated widespread anxiety about the future of work, technological unemployment, and economic disruption, with some early evidence showing large productivity gains (Brynjolfsson, Li and Raymond, 2025; Noy and Zhang, 2023).

This uncertainty reflects a potential shift in the nature of this technological innovation. Unlike previous technologies that predominantly displaced routine tasks in lower- and middle-skilled occupations (Autor, 2015; Autor et al., 2003), LLMs possess capabilities that extend across the entire skill distribution, with the potential to affect highly skilled professional and analytical roles as well (Eloundou et al., 2024). Estimates suggest that around 80% of the U.S. workforce could see at least 10% of their tasks affected, with exposure rising in higher-income occupations on average (Eloundou et al., 2024). As such, the introduction of generative AI raises important questions about whether LLMs will erode job security, depress white-collar wages, and increase or decrease inequality.

To examine these questions, we use detailed UK labor market data from Revelio Labs, covering millions of firms and job postings between 2021 and 2025. Our identification strategy exploits the sudden and arguably exogenous introduction of ChatGPT in November 2022. The impact of this shock varied across firms and industries depending on their occupational mix and the extent to which core activities are susceptible to LLM automation. We measure this heterogeneity using the occupational level LLM exposure scores developed by Eloundou et al. (2024), who evaluate 19,265 detailed work activities from the O*NET database using both human annotators and GPT-4 to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. These exposure scores allow us to quantify the share of tasks within each occupation, or the share of jobs within each firm that can be performed by LLMs.¹

¹Their measure strongly correlates with other LLM exposure measures (see e.g. Tomlinson et al., 2025; Hampole et al., 2025).

Building on these exposure measures, we estimate a series of difference-in-differences models around the introduction of ChatGPT that exploit firm- and occupation-level variation in LLM exposure to estimate the causal effects of the ChatGPT shock on employment, wages, pay dispersion, job seniority, workplace arrangements, and overall hiring dynamics.

Our results show a clear impact of LLM introduction on UK labor markets, with effects emerging gradually across multiple dimensions of employment. At the firm level, a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment. The impact is concentrated among junior positions, with low-seniority employment declining by 0.4 percent while high-seniority positions show smaller and partly insignificant reductions. This compositional shift reduces the fraction of low-seniority workers, which mechanically raises average compensation in exposed companies as the workforce skews toward higher-paid senior employees.

These patterns also show up in hiring intentions. In an analysis of firm-level job posting data, we find that a one standard deviation increase in LLM exposure reduces the likelihood that firms post job vacancies in a given month by 1.0 percentage point. The reduction in job postings among exposed firms is strongest for technical and creative roles such as software engineers and designers. At the occupation level, we observe even more pronounced effects on hiring. A one standard deviation increase in an occupation’s LLM exposure is associated with a 3.9 percent decline in job postings, accompanied by substantial reductions in posted salaries.

Importantly, these effects are not uniformly distributed across the labor market. Our heterogeneity analysis shows that LLM disruption is concentrated almost entirely in higher-wage segments. High-compensation firms experience employment reductions of 0.7 percent per standard deviation increase in exposure, compared with near-zero effects for low-compensation firms. Similarly, high-salary occupations experience 5.7 percent declines in job postings per standard deviation, while low-salary roles show no significant change. These results point in the opposite direction from historical automation experiences, in which routine middle-skill occupations bore the brunt of technological displacement (Autor, 2024).

To interpret these results, we draw on the task-based framework (Autor et al., 2003; Acemoglu and Restrepo, 2018, 2019, 2020; Autor and Thompson, 2025). This framework models production as discrete tasks allocated between labor and capi-

tal based on comparative advantage. When automation enables capital to perform tasks previously done by labor more cheaply, it creates a *displacement effect* that reduces labor demand. Conversely, the creation of new tasks where labor maintains comparative advantage generates a *reinstatement effect* that increases labor demand (Acemoglu and Restrepo, 2019; Restrepo, 2024). For LLMs, which can automate cognitive tasks previously considered non-routine, the framework predicts that displacement will dominate in the short run, particularly for entry-level positions involving routine information processing and basic analysis. Senior roles requiring tacit knowledge, judgment, and complex interpersonal coordination should prove less vulnerable. This is consistent with our findings.

Our findings contribute to the emerging international evidence on labor market impacts of AI (del Rio-Chanona et al., 2025). In the United States, Brynjolfsson, Chandar and Chen (2025); Lichtinger and Hosseini Maasoum (2025); Dominski and Lee (2025) document employment declines in exposed industries concentrated in early-career positions. However, Chandar (2025) finds no employment effects using CPS data, and Humlum and Vestergaard (2025) report null results from Danish administrative data. These disparities may reflect several factors: differences in labor market institutions between countries, variation in the timing of LLM adoption across regions, and methodological choices regarding exposure measurement and outcome variables. Our UK evidence suggests that effects manifest differently across adjustment margins, with hiring responding more quickly and strongly than employment stocks.

Methodologically, our paper builds on recent work that also uses Revelio Labs data to construct firm-level AI exposure measures. Eisfeldt et al. (2023) show that higher-exposure US firms experience excess stock returns following the release of ChatGPT, while Hampole et al. (2025) develop task-level exposure measures varying across firms and occupations to study US labor demand between 2010 and 2023. We extend their work in three directions: we shift focus to the UK labor market, examine realized employment and job posting outcomes (as opposed to Eisfeldt et al. (2023)’s focus on financial data), and exploit data through mid-2025 to capture labor market adjustment to generative AI adoption (as opposed to Hampole et al. (2025)’s focus on earlier waves of AI).

In the UK context, we extend Henseke et al. (2025), who develop a Generative AI Susceptibility Index and provide suggestive evidence that wage premia decline

in exposed occupations. While their analysis covers 25 broad occupation groups, we exploit variation across a much wider variety of job types, as well as a firm-level analysis that examines within-firm dynamics that would be invisible in occupational aggregates. In addition, our individual-level posting data enables examination of a wider set of outcome variables such as salary distributions and seniority-specific employment, as well as heterogeneity in treatment effects by pre-treatment compensation or location.

Taken together, our findings raise important policy questions. The disproportionate elimination of entry-level positions disrupts the conventional model of skill development, where employees master progressively complex tasks through hands-on experience with routine work. Without junior roles that serve as training grounds, firms may struggle to develop future senior talent internally, creating a hollowed-out labor market where entry becomes increasingly difficult (Ide, 2025). This dynamic could exacerbate inequality as those with pre-existing experience benefit from reduced competition while new entrants face shrinking opportunities. Policy responses must address both immediate displacement and longer-term skill development pathways.

2 Data

Our primary data comes from Revelio Labs, which aggregates and processes publicly available employment information from professional networking platforms, company websites, and job boards using machine learning algorithms to create proprietary longitudinal labor market databases. Revelio Labs transforms semi-structured public profile information into a structured panel dataset through systematic parsing, standardization, and cleaning procedures. Their data have been validated against official statistics and used in recent academic research on labor market dynamics (Fadhel et al., 2021; Eisfeldt et al., 2023; Hampole et al., 2025). In Appendix A4, we separately validate that Revelio Labs employment and job posting data closely track Office for National Statistics aggregates both over time and across industries. We use two separate Revelio Labs datasets: work profiles and job postings.²

²The data used in this paper were downloaded on 12 August 2025.

2.1 Job Profile Data

The first dataset consists of all publicly available professional LinkedIn profiles between January 2021 through June 2025. The dataset includes approximately 75 million employment spells from 23 million unique individuals across 2.5 million UK-based firms. Each observation includes the worker’s job title, employer, start and end dates (if available), seniority level,³ and compensation.⁴ The data provide near-real-time employment stocks, which enable us to track within-firm workforce composition changes over time.

From the panel of individual LinkedIn profiles, we construct a monthly firm-level panel from January 2021 through June 2025. For each firm-month, we calculate total employment as the count of unique individuals with active employment spells at a certain company in that month, and log-transform this variable. This is our main employment stock measure. We then separate total employment into junior and senior employment stocks based on Revelio seniority classifications of jobs. We also calculate average compensation as the log of the mean of reported salaries plus bonuses in a firm. We only focus on firms that remained in business throughout the sample period.

To calculate firm-level exposure scores, we use the assessment framework developed by Eloundou et al. (2024). They evaluate 19,265 detailed work activities and tasks from the O*NET 27.2 database to determine whether LLM assistance could reduce task completion time by at least 50% while maintaining quality. We focus on their ‘E1’ measure, which captures exposure to current LLMs, rather than LLMs combined with additional software. We do so because it is the most appropriate metric for identifying short-term labor market effects that do not require additional software development or complex organizational changes. To obtain our firm-level

³Revelio Labs determines seniority using an ensemble model that combines three components: (i) current job characteristics including title, company, and industry; (ii) employment history including tenure and previous position seniority; and (iii) individual age. These three scores are averaged to create a continuous seniority metric, which is then mapped to seven ordinal levels based on keyword benchmarks (e.g., “junior,” “senior,” “director”). The levels range from Entry Level (e.g., interns, trainees) through Junior, Associate, Manager, and Director levels, to Executive and Senior Executive levels (e.g., C-suite positions).

⁴Revelio Labs estimates compensation using a predictive model trained on publicly available visa application data, self-reported salaries, and job posting information. The model incorporates position-specific variables (job title, seniority level, company, location), individual-specific factors (tenure at company), and temporal controls (year of observation) to generate salary predictions for positions where compensation data are not directly observed.

measures, we first obtain occupation-level measures by calculating the share of tasks within an occupation that meet the 50% time-saving threshold. We weight this by task importance, assigning 0.5 to supplementary tasks and 1.0 to core tasks.⁵

Formally, for occupation o :

$$\alpha_o = \sum_{t \in T_o} w_t \cdot \mathbb{1}_{[\text{LLM reduces time by } \geq 50\%]} \quad (1)$$

where T_o denotes occupation o 's task set, w_t represents task importance weights (1.0 for core tasks, 0.5 for supplemental), and $\mathbb{1}_{[\cdot]}$ is an indicator function for substantial LLM impact. α_o gives us an occupation-level exposure score that we can assign to each employee.

We then calculate each firm's average exposure score by taking the average exposure score of all employees who report working for that firm on 30 October 2022, immediately before the introduction of ChatGPT.⁶

Formally:

$$\text{Exposure}_i = \sum_o \frac{L_{io, \text{Oct2022}}}{L_{i, \text{Oct2022}}} \cdot \alpha_o \quad (2)$$

where $L_{io, \text{Oct2022}}$ denotes firm i 's employment in occupation o as of October 2022.

We restrict our sample of firms to those with at least 100 employees immediately before ChatGPT's introduction to address two methodological concerns. First, larger firms reduce noise from idiosyncratic employment changes and ensure adequate LinkedIn coverage for constructing reliable firm-level exposure scores from individual employee profiles. Second, this threshold minimizes attrition bias from firms exiting the sample while maintaining a sufficiently large set of employers that account for a large part of UK employment. Appendix A3.3 in the Appendix examines the sensitivity of our results to different firm-size thresholds.

Our final firm-level sample includes 26,979 firms with 1,268,013 firm-month ob-

⁵Our robustness checks cover alternative weights based on O*NET task-importance classifications.

⁶We measure firm-level exposure using the pre-treatment workforce composition to avoid endogeneity concerns. If we calculated exposure using post-ChatGPT employment, our measure would be contaminated by the treatment effect itself, as firms might have already adjusted their occupational mix in response to LLM capabilities. By fixing exposure at the pre-treatment date, we ensure that our exposure measure captures firms' initial vulnerability to LLM disruption rather than their endogenous response to it.

Table 1: Summary Statistics Firm Employment

	Low exposure		High exposure	
	Before	After	Before	After
Total jobs	456	507	639	708
Low seniority (%)	0.705	0.703	0.667	0.662
High seniority (%)	0.295	0.297	0.333	0.338
Compensation	£ 94,325.66	£ 94,079.89	£100,230.08	£100,184.41
LLM exposure score	0.096	0.096	0.172	0.172

Notes: This table reports summary statistics on firm-level employment outcomes by exposure to large language models (LLMs). Firms are classified into high- and low-exposure groups based on whether their average exposure score lies above or below the median across all firms. *Total jobs* gives the mean number of active positions in the firm. *Low seniority* and *High seniority* denote the share of employees in junior and senior roles, respectively, as defined by seniority of 1-3 (junior) or 4-7 (senior). *Compensation* gives the mean annual salary plus bonuses, and *LLM exposure score* is the average predicted LLM-exposure of jobs in a firm based on Eloundou et al. (2024). *Before* refers to the pre-treatment period (prior to November 2022), while *After* refers to the post-treatment period (November 2022 onward).

servations. These firms employ 16 million workers in our data as of October 2022, representing approximately 49% of UK employment (Office for National Statistics, 2022).

Table 1 presents firm-level employment summary statistics. Firm-level employment stocks increased for both exposure groups following ChatGPT’s introduction, rising 11.2 percent for low-exposure firms (from 456 to 507 employees) and 10.8 percent for high-exposure firms (from 639 to 708 employees). However, this increase could reflect an increase in the number of people joining LinkedIn during the sample period. The workforce composition reveals important differences: high-exposure firms maintain a more senior-skewed workforce, with 33.3 percent of positions at high-seniority levels compared to 29.5 percent in low-exposure firms. Following ChatGPT’s release, this seniority gap widened slightly, as the share of high-seniority positions increased by 0.5 percentage points in high-exposure firms versus 0.2 percentage points in low-exposure firms. These patterns suggest that high-exposure firms are gradually shifting their workforce composition toward more senior roles, potentially reflecting the displacement of entry-level positions. Consistent with their more senior workforce composition, high-exposure firms offer higher average compensation.

2.2 Job Listing Data

The second Revelio Labs dataset contains almost the entire universe of online job postings in the UK over the same period, 63,497,648 in total. The data are collected from vacancies from major job boards, company career pages, and recruitment platforms. Each posting includes the job title, employer, posting and removal dates, salary information, and full job ads. This job listing data captures labor demand flows and allows us to observe how hiring intentions and job characteristics evolve. We use this dataset for two purposes: to examine how hiring intentions evolve across exposed and less-exposed firms and across exposed and less-exposed occupations.

To construct our firm-level hiring intentions data, we use the same sample of firms with at least 100 employees in October 2022 used in our total employment analysis, but focus on flows (job listings) instead of employment stocks. We create a monthly panel at the firm level, tracking job posting activity from January 2021 through June 2025. Given that approximately 50% of firms do not post job listings in any given month, we construct a binary outcome variable indicating whether a firm posted any job listings in a given month, rather than using posting counts. This measure captures the extensive margin of hiring activity. We again match each firm to its pre-treatment LLM exposure score calculated from the October 2022 workforce composition, as described above. We also break postings down by job roles (50 in total) to study which types of roles exposed firms hire more or less of.

This firm-level posting dataset allows us to test whether firms with higher LLM exposure adjust their hiring intensity differently than less-exposed firms following ChatGPT’s introduction, providing insights into how anticipated productivity gains or workforce restructuring needs translate into recruitment behavior. In addition, by examining the occupational composition of job postings within firms, we can identify whether highly exposed firms disproportionately reduce hiring for specific types of roles (for example, those with high LLM exposure scores) relative to less exposed firms, allowing us to test for within-firm occupational reallocation in response to LLM adoption.

To construct our occupation-level hiring intentions data, we first match each vacancy to O*NET SOC occupation codes using a job-role-SOC crosswalk provided by Revelio Labs. This crosswalk maps each unique role (1,500 in total) to 384 broader categories that directly correspond to the occupations listed in Eloundou et al.

Table 2: Summary Statistics Occupation-Level Job Postings

	Low exposure		High exposure	
	Before	After	Before	After
Job postings	4,056	3,358	3,343	2,745
Salary	£44,828.34	£49,359.93	£46,177.33	£50,262.62
Salary (SD)	£13,940.86	£15,238.06	£13,469.71	£14,136.39
LLM exposure score	0.045	0.045	0.263	0.263

Notes: This table reports occupation-level summary statistics on job postings by exposure to large language models (LLMs). Occupations are classified into high- and low-exposure groups based on whether their exposure score lies above or below the median score. *Job postings* gives the mean number of vacancies posted. *Salary* reports the mean advertised annual compensation, while *Salary (SD)* gives the corresponding standard deviation across postings. *LLM exposure score* is the average predicted LLM-exposure of tasks in the occupation, based on Eloundou et al. (2024). *Before* refers to the pre-treatment period (prior to November 2022), while *After* refers to the post-treatment period (November 2022 onward).

(2024). We then aggregate vacancies to the occupation-month level. Our monthly occupation-level variables capture multiple dimensions of labor demand. Posting volume is measured as the total new vacancies posted, which we log-transform. Posted salary represents the mean advertised salary for positions (level or log).⁷ We additionally calculate salary dispersion as the standard deviation of posted salaries within occupation-month cells to capture within-occupation inequality effects.

Table 2 presents summary statistics for occupation-level job postings. We separate the data by high vs. low exposure occupations, and show the data for the periods before and after the release of ChatGPT in November 2022. Job posting volumes declined across both exposure groups following the introduction of ChatGPT, with high-exposure occupations experiencing a sharper decline of 17.9 percent (from 3,343 to 2,745 monthly postings) compared to 17.2 percent for low-exposure occupations (from 4,056 to 3,358). Second, average posted salaries increased for both groups, rising from £44,828 to £49,360 (10.1 percent) in low-exposure occupations and from £46,177 to £50,263 (8.8 percent) in high-exposure occupations, with salary dispersion increasing similarly across both groups. Finally, high-exposure occupations have LLM exposure scores approximately 6 times higher than low-exposure occupations.

⁷When salary is not explicitly listed, it is inferred from a predictive model based on job characteristics, company, and location, trained on visa applications, self-reported data, and job postings, with salaries inferred for companies lacking direct data.

3 Methodology

We use a difference-in-differences design with continuous treatment intensity to estimate the effect of LLM exposure on UK labor market outcomes. Our identification strategy exploits variation in occupational and firm-level exposure to LLM capabilities, as measured by Eloundou et al. (2024), combined with the timing of ChatGPT’s public release in November 2022. We conduct three types of analyses: firm-level employment stocks using LinkedIn profile data, firm-level hiring decisions based on job posting data, and occupation-level hiring patterns using job posting data.

For our firm-level employment stock analysis, we estimate the following specification:

$$Y_{it} = \sum_{\tau=-15}^{30} \beta_{\tau} \cdot \text{Exposure}_i \times \mathbb{1}_{[t=\tau]} + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (3)$$

where Y_{it} represents outcomes for firm i in month t , including log total employment, log employment by seniority level (junior vs. senior), the fraction of low-seniority positions, and average compensation. Exposure_i is the firm’s pre-treatment LLM exposure score, calculated as the employment-weighted average of occupational exposure scores using the workforce composition on 30 October. We standardize the exposure coefficients across all analyses to make the coefficients more easily interpretable. The term $\mathbb{1}_{[t=\tau]}$ denotes indicators for event time τ relative to November 2022, with October 2022 ($\tau = -1$) as the reference period. We include firm fixed effects γ_i to control for time-invariant firm characteristics, and industry-by-period fixed effects δ_{jt} to account for sector-specific shocks that might correlate with technology adoption patterns. Standard errors are clustered at the firm level.

This event study design allows us to test for parallel pre-treatment trends, as well as to examine the evolution of treatment effects over time, showing when and how labor market impacts start to emerge, and whether they persist, intensify, or attenuate over time. We use 15 months of pre-treatment data and 30 months of post-treatment data. To summarize the overall treatment effect, we also report average post-treatment effects from a static specification:

$$Y_{it} = \beta \cdot \text{Exposure}_i \times \text{Post}_t + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (4)$$

where Post_t is an indicator for periods after November 2022.

To examine how firms adjust their hiring strategies in response to LLM exposure, we combine our firm-level exposure measures with job posting data. To do so, we estimate Equation (3) with an indicator for whether firm i posted any job listings in month t as the outcome variable. This binary outcome captures the extensive margin of total hiring activity. We then re-estimate Equation (4) separately for each of 50 job categories.

For our occupation-level analysis of job posting dynamics, we estimate an analogous event study specification:

$$Y_{ot} = \sum_{\tau=-15}^{30} \beta_{\tau} \cdot \text{Exposure}_o \times \mathbb{1}_{[t=\tau]} + \gamma_o + \delta_t + \epsilon_{ot} \quad (5)$$

where Y_{ot} denotes outcomes for occupation o in month t , including log job posting volumes, posted salaries, and salary dispersion within occupations. Exposure_o is the occupation’s LLM exposure score from Eloundou et al. (2024). We include occupation fixed effects γ_o and year-by-month fixed effects δ_t with standard errors clustered at the occupation level. We also estimate an analogous static specification of Equation (4) at the occupation level.

Our identification strategy relies on the parallel trends assumption: absent ChatGPT’s introduction, labor market outcomes would have evolved similarly across firms and occupations with different LLM exposure levels, conditional on our fixed effects structure. Several features of our design help with the interpretation of our estimates as causal. First, our exposure measures are predetermined using data from before November 2022, eliminating concerns about reverse causality where employment changes might influence measured exposure. Second, the sudden and largely unanticipated public release of ChatGPT provides a clear temporal shock to LLM availability. Third, our event study analyses test and confirm parallel pre-treatment trends across analyses.

Furthermore, we perform a large set of robustness checks to test the validity of our findings across different specifications and modeling choices. We re-estimate our analyses using different fixed effects. We examine the validity of our finding of decreased junior hiring through a highly restrictive within-firm triple-difference model using a firm-by-month-by-seniority panel that includes firm-by-period fixed effects absorbing all time-varying firm-level shocks. We test the sensitivity of our results to

sample construction by varying minimum firm size thresholds. We consider alternative task-level weights to construct our exposure measures. We estimate weighted regressions that give greater weight to larger firms and occupations to ensure our results reflect typical worker experiences. Finally, we relax the assumption of linear treatment effects by categorizing firms and occupations into exposure quartiles. These analyses consistently support our main findings.

These robustness checks notwithstanding, we acknowledge the possibility that other factors differentially affected exposed and less exposed firms or occupations around the time of ChatGPT’s introduction. For instance, if firms anticipating AI disruption had already begun adjusting their workforce composition before November 2022, or if other technological or economic shocks coincidentally affected high-exposure entities differently, our estimates could be biased. Additionally, our exposure measures capture potential susceptibility to LLM disruption but may not perfectly predict actual adoption patterns, which likely vary with firm-specific factors such as technological capabilities, management quality, and financial resources.

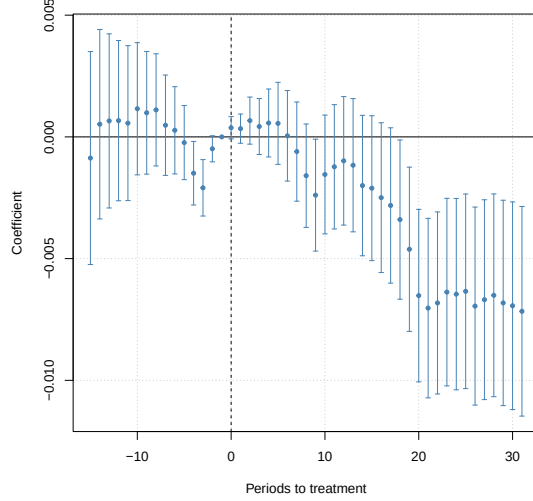
4 Results

This section presents our main empirical findings on the effects of LLM exposure on UK labor market outcomes. We organize the analysis in three parts. First, we examine how employment stocks, workforce composition, and compensation evolve within firms with different levels of LLM exposure. Second, we investigate occupation-level hiring patterns through job posting data. Third, we explore within-firm recruitment strategies by combining firm-level exposure measures with their occupation-specific hiring patterns. Throughout, we present both event study specifications and static difference-in-differences estimates.

4.1 Firm-Level Employment Effects

Figure 1 presents the event study plots for the effect of LLM exposure on the total number of employees in UK firms. We find that highly exposed firms start experiencing a statistically significant decline in total employment beginning approximately 18 months after the introduction of ChatGPT, suggesting that exposed firms reduce headcount in response to this new technology, but that it takes time to

Figure 1: Effect of LLM exposure on firm-level jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the total number of employees in UK firms (log). Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. Vertical line indicates November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

show up in aggregate employment statistics. High and low LLM exposure exhibit similar employment trajectories prior to November 2022, which provides support for the validity of the parallel trends assumption. When averaged over the entire post-treatment period (Table 3), a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment, although the aggregate effect is only marginally significant due to the delayed onset of impacts.

We next examine how these aggregate effects vary across different types of workers. Figure 2 decomposes the effects between high- and low-seniority employees. Panel A shows junior positions decline immediately and statistically significantly, while Panel B demonstrates that senior positions initially remain stable before declining gradually, with the decrease only becoming significant after 20 months. Panel C shows that the timing difference in the emergence of effects leads to a temporary shift toward senior employees, although the compositional change attenuates over time, with the seniority distribution gradually reverting toward pre-treatment levels by the end of our observation period. All three panels show parallel pre-trends.

Table 3: Effect of LLM exposure on firm-level job outcomes

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.003* (0.002)	−0.004** (0.002)	−0.002 (0.002)	−0.001** (0.0003)	106.313* (59.029)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, averaged over the post-treatment period. *Employees* is the total number of jobs in the firm. *Low seniority* and *High seniority* denote the number of positions in seniority levels 1–3 and 4–7, respectively. *Low seniority (%)* is the share of jobs that are low-seniority. *Av. comp.* is the mean annual salary plus bonuses. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level and reported in parentheses. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regression results in Table 3 show, averaged over the entire post-treatment period, a significant 0.4 percent reduction in junior employees per standard deviation increase in exposure, and a statistically insignificant reduction in senior employees. The fraction of junior employees goes down by 0.1 percentage points per standard deviation of LLM exposure ($p = 0.028$).

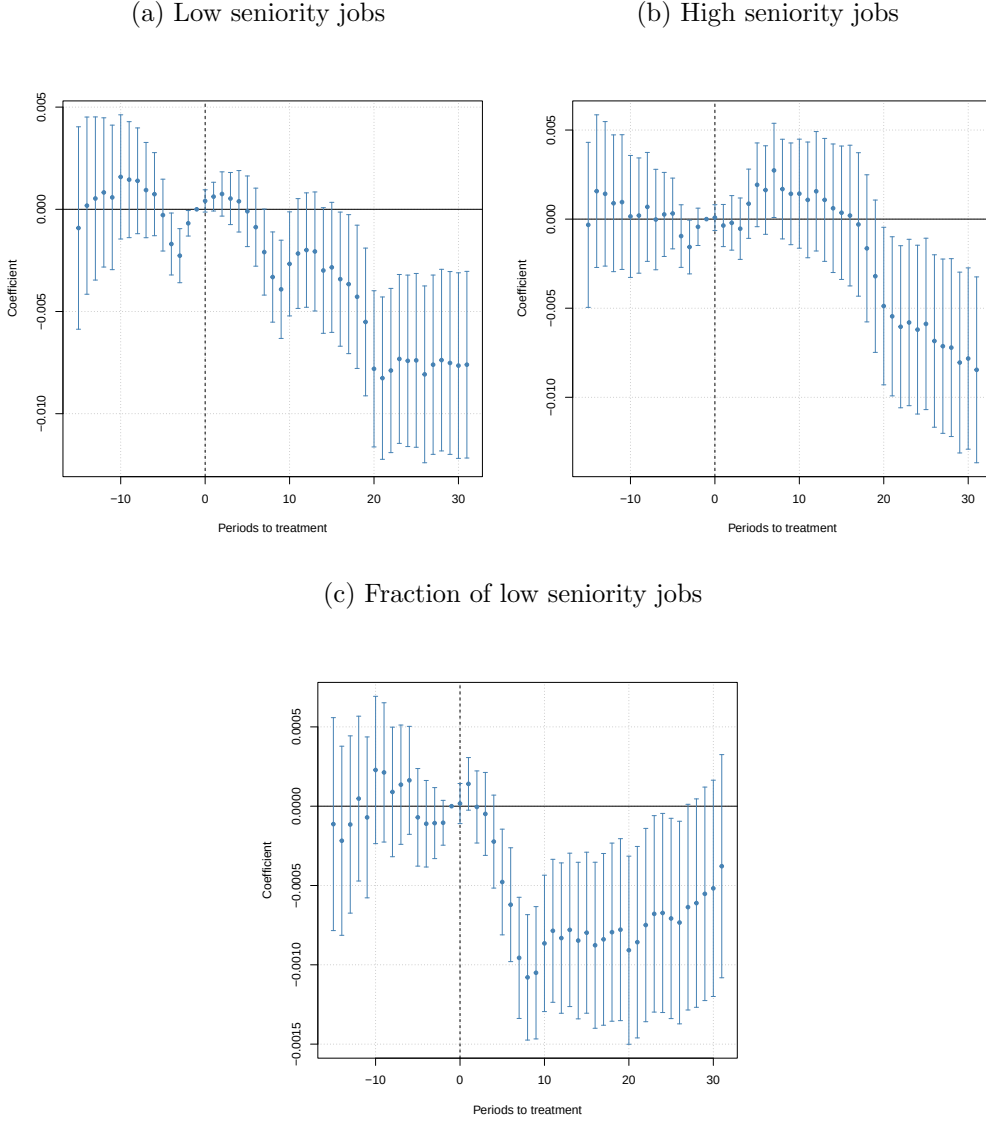
The observed compositional shift towards more senior roles has direct implications for compensation patterns. Figure 3 shows that average compensation rises modestly in highly exposed firms following the introduction of ChatGPT, with a one standard deviation increase in exposure associated with a £106 increase in average compensation, although the overall estimate is only marginally significant.

4.2 Job Posting Dynamics

4.2.1 Firm-Level Postings

While employment stocks adjust slowly, job postings provide a real-time window into firms’ evolving labor demand. We therefore examine how firms’ hiring intentions respond to LLM exposure by analyzing whether exposed firms post fewer vacancies overall and whether they adjust the composition of roles they seek to fill. This posting data captures the extensive margin of recruitment activity and reveals forward-looking decisions that precede the employment stock changes documented above. By tracking the same firms across both datasets, we can directly observe

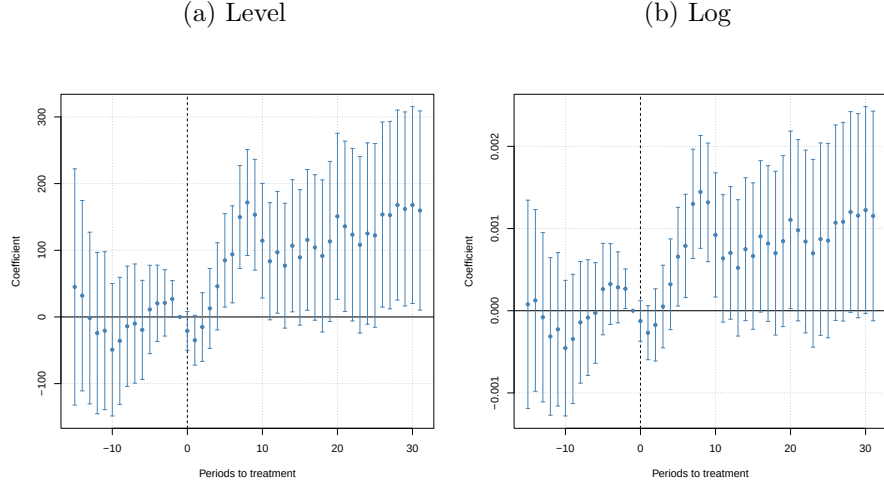
Figure 2: Effect of LLM exposure on firm-level seniority of jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the seniority composition of jobs within UK firms. Panel A shows effects on the total number of low-seniority positions (seniority levels 1–3), panel B on the total number of high-seniority positions (levels 4–7), and panel C on the fraction of jobs that are low-seniority. Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

how hiring intentions translate into gradual changes in employment.

Figure 3: Effect of LLM exposure on firm-level compensation



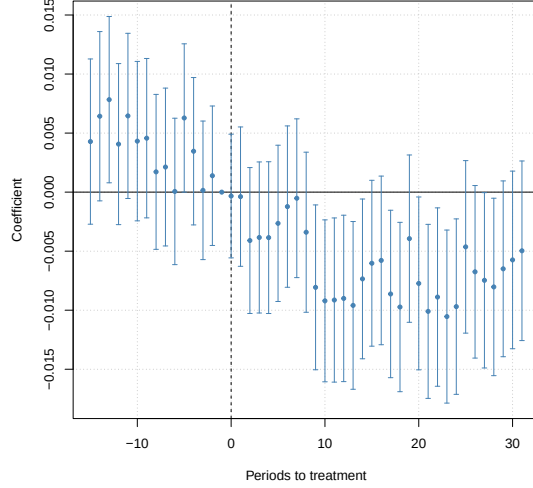
Notes: This figure plots event study estimates of the effect of LLM exposure on average compensation within UK firms. Compensation is defined as annual base salary plus bonuses. Panel A shows compensation in levels, Panel B in logs. Coefficients are estimated from Equation (3) and are shown relative to October 2022, the final pre-treatment month. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

Figure 4 presents event study estimates for the probability that exposed vs. less exposed firms post any job vacancy in a given month. The results show a change in hiring patterns following November 2022. Prior to this date, firms with different exposure levels did not have statistically significantly different trends in job postings.⁸ Following the release of ChatGPT, however, we observe that a one standard deviation increase in LLM exposure reduces firms' likelihood of posting new jobs by 1.0 percentage point ($p < 0.001$, see Table A1). This effect emerges gradually and becomes statistically significant after approximately 10 months, mirroring the timeline of employment stock adjustments.

Next, we decompose the aggregate hiring reduction by estimating occupation-specific effects across job categories. We estimate Equation (4) separately for 50 distinct occupational categories, with the outcome variable indicating whether a

⁸Although all pre-trend estimates are statistically insignificant, the point estimates exhibit a slight downward drift. If this reflects a differential pre-trend rather than sampling variation, the post-treatment estimates may overstate the true effect.

Figure 4: Effect of LLM exposure on firm-level job listings

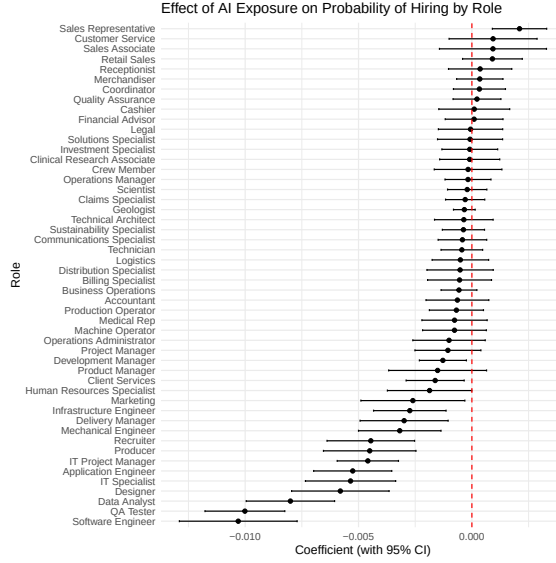


Notes: This figure plots event study estimates of the effect of LLM exposure on the probability that a UK firm posts any job vacancy in a given month. Coefficients are estimated from Equation (3) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable is an indicator equal to one if the firm posted at least one vacancy in month t . The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

firm posts any vacancy for that specific role in a given month. We continue to compare exposed vs. less exposed firms in the same time period.

Figure 5 presents these occupation-specific treatment effects, ordered by magnitude. The results reveal substantial heterogeneity in hiring adjustments across roles. Technical positions experience the sharpest declines: a one standard deviation increase in firm-level LLM exposure reduces the likelihood of posting software engineer positions by 1.1 percentage points ($p < 0.001$), QA tester positions by 1.0 percentage points ($p < 0.001$), and data analyst positions by 0.8 percentage points ($p < 0.001$). We find similar declines for other technical roles such as IT specialists, IT project managers, and infrastructure engineers. Creative and design roles also see significant reductions, with job postings for designer positions decreasing by 0.6 percentage points ($p < 0.001$) and for producers by 0.5 percentage points ($p < 0.001$). In contrast, sales representative positions show a statistically significant increase of 0.3 percentage points ($p < 0.001$). This pattern suggests that exposed firms are scaling back recruitment for technical and analytical roles where LLMs can augment or

Figure 5: Effect of LLM exposure on firm-level hiring across job categories



Notes: This figure reports coefficient estimates and 95% confidence intervals for the effect of LLM exposure on firms' hiring at the extensive margin across 50 job roles. For each role, we estimate Equation (4) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable equals one if a firm posts at least one vacancy for that job role in month t .

replace human capabilities, while maintaining or slightly increasing hiring for roles requiring direct customer interaction and relationship building that remain beyond current LLM capabilities.

Importantly, to emphasize that these estimates capture differential responses between exposed and less exposed firms, not absolute changes in hiring patterns. Hence, a coefficient of zero does not imply that hiring for a particular role remained unchanged. Rather, it indicates that exposed and less exposed firms adjusted their demand for that role similarly. For instance, if all firms reduced hiring of software engineers, regardless of their own exposure levels, our estimates would show no differential effect. The significant negative coefficient for e.g. data analysts therefore reveals that exposed firms reduced analyst hiring beyond any economy-wide reduction, suggesting that firm-level exposure amplifies the substitution effects already occurring across the labor market. We next turn to our occupation-level analysis to examine how hiring intentions of exposed jobs change in the aggregate.

Table 4: Effect of LLM exposure on occupation-level job postings

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.039** (0.018)	-495.483*** (135.270)	-0.011*** (0.003)	-56.802 (108.358)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.953	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation o in month t . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

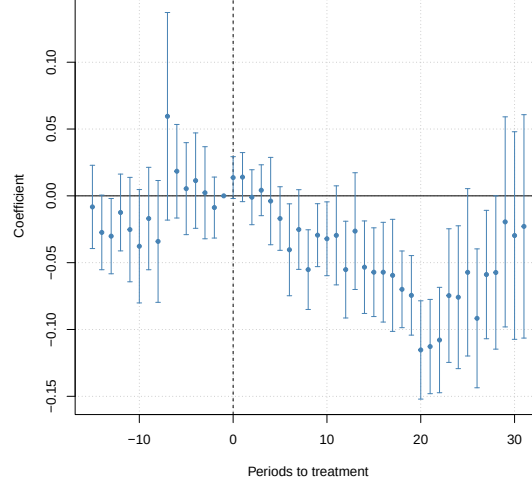
4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ($p = 0.029$). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log

Figure 6: Effect of LLM exposure on job-level hiring

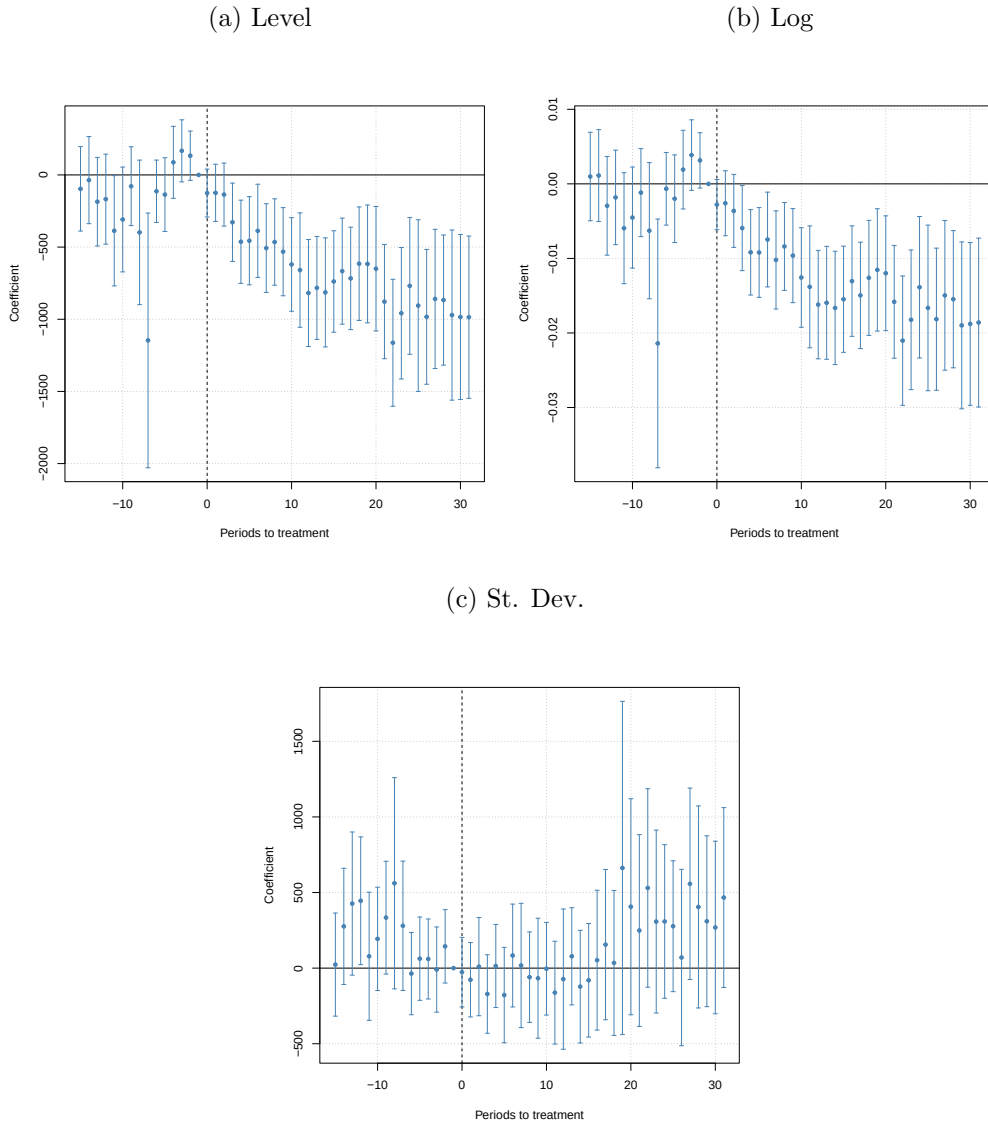


Notes: This figure plots event study estimates of the effect of LLM exposure on occupation-level hiring (log vacancies per occupation-month). Coefficients are estimated from Equation (5) with occupation fixed effects and year-by-month fixed effects; standard errors are clustered at the occupation level. The dependent variable is the log of the number of new vacancies posted in occupation o in month t . Exposure is the occupation-level score from Eloundou et al. (2024). The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

salaries (both $p < 0.001$). Panel C shows that within-occupation salary dispersion remains unchanged, suggesting uniform wage pressure across the distribution rather than polarization between high- and low-earners within occupations.

These occupation-level salary declines provide a sharp contrast with the slight increase in firm-level compensation documented earlier. At the firm level, eliminating junior positions mechanically raises average compensation by shifting workforce composition toward senior roles. At the occupation level, however, we observe market-wide wage depression for exposed roles as increased labor supply (from displaced workers) meets reduced demand. As such, our results highlight the importance of considering both within-firm and within-occupation changes.

Figure 7: Effect of LLM exposure on occupation-level job-posting salary



Notes: This figure plots event study estimates of the effect of occupation-level LLM exposure on salaries in job listings. The dependent variables are posted salary per job listing (Panel A), log salary (Panel B), and the standard deviation of the posted salary (Panel C). All other definitions are as in Figure 6.

4.3 Heterogeneous Treatment Effects

Our main results show the average treatment effects across all firms and occupations, but the impact of LLM exposure is likely to vary with organizational and labor mar-

ket characteristics. We examine heterogeneity along two dimensions: compensation levels (splitting firms and occupations by median pre-treatment salary) and geographic concentration (comparing London-based versus non-London entities). These dimensions are highly policy-relevant, as compensation-level heterogeneity provides a first indication as to whether LLMs may compress or widen wage inequality, while geographic variation tests whether disruption concentrates in London’s knowledge economy or extends throughout the UK labor market.

4.3.1 Heterogeneity by Compensation Level

Table A2 examines how firm-level employment effects vary by average compensation, with high-compensation jobs serving as a proxy for more highly educated white collar work, and low-compensation jobs for relatively low-educated blue collar work. We split firms at the median pre-treatment compensation level. Panel A shows that high-compensation firms experience substantially larger employment reductions than their lower-paying counterparts. A one standard deviation increase in LLM exposure reduces total employment at high-compensation firms by 0.7 percent ($p = 0.009$), driven primarily by a 0.8 percent reduction in low-seniority positions ($p = 0.005$). In contrast, Panel B indicates that low-compensation firms show no significant employment effects, with point estimates near zero for both total employment and low-seniority positions.

This pattern extends to hiring intentions. Table A4 shows that a one standard deviation increase in LLM exposure reduces the probability of posting vacancies by 1.3 percentage points at high-compensation firms ($p < 0.001$), compared to a 0.7 percentage point reduction at low-compensation firms ($p = 0.001$). While both groups curtail hiring, the effect is approximately twice as large at high-compensation firms.

The occupation-level analysis in Table A5 provides further evidence that the employment effects are concentrated among the highly paid. In high-salary occupations, a one standard deviation increase in LLM exposure is associated with a 5.7 percent reduction in job postings ($p = 0.009$) and £782 salary declines ($p < 0.001$), while low-salary occupations show no significant changes in posting volumes and minimal salary effects.

These results suggest that LLM disruption is almost entirely driven by changes

in higher-wage segments of the labor market, contradicting historical automation patterns that predominantly affected routine middle-skill occupations. The concentration of employment losses and wage pressures in high-compensation firms and high-salary occupations implies that LLMs may have an equalizing effect on the income distribution, at least in the short run. This pattern aligns partially with the optimistic scenario outlined by Autor (2024) who argues that AI could help rebuild middle-class jobs by enabling workers with foundational expertise to perform higher-stakes decision-making tasks previously reserved for elite professionals. However, our findings suggest a more pessimistic mechanism: rather than elevating middle-skill workers to perform expert tasks, LLMs appear to be eroding demand for the expert tasks themselves. While this may compress wage inequality by reducing returns to education and expertise, it appears to achieve this through a leveling-down rather than the leveling-up process envisioned by Autor (2024).

4.3.2 Geographic Heterogeneity

Geographic concentration provides another lens for understanding differential treatment effects. Table A3 compares firms headquartered in London versus elsewhere in the UK. Surprisingly, London-based firms show minimal employment responses despite their concentration in LLM-exposed industries like finance and professional services. Total employment effects are statistically insignificant (0.1 percent per standard deviation), with similar null results for both seniority levels. In contrast, there is some evidence that firms headquartered outside London experience stronger employment declines from LLM exposure: a one standard deviation increase in LLM exposure reduces total employment by 0.4 percent ($p = 0.042$), with low-seniority positions declining by 0.5 percent ($p = 0.020$). That said, the firm-level hiring response shows the opposite pattern: a one standard deviation increase in LLM exposure reduces job postings by 1.6 percentage points for London-based firms versus 0.8 percentage points for firms outside London, despite their more modest employment stock adjustments.

At the occupation level, both London and non-London job markets experience significant disruption. A one standard deviation increase in LLM exposure is associated with a 3.4 percent reduction in job postings in London ($p = 0.047$) with salary declines of £563 ($p < 0.001$), while positions outside London experience posting

reductions of 4.0 percent ($p = 0.030$) and salary declines of £476 ($p < 0.001$).

4.4 Robustness Checks

We conduct several robustness tests to assess the sensitivity of our results to alternative specifications and sample restrictions. First, we examine whether our findings depend on the fixed effects structure. Tables A7 and A8 present firm-level results using year-by-month fixed effects instead of industry-by-year-by-month fixed effects, while Table A9 replicates our occupation-level analysis using job category-by-period fixed effects. Despite these alternative specifications, all main conclusions persist: low-seniority employment declines at exposed firms, hiring decreases are, if anything larger, and occupation-level job postings and salaries fall significantly with exposure.

Second, we go one step further to address potential endogeneity concerns, and implement an extremely restrictive within-firm specification. To do so, we construct a firm-by-month-by-seniority panel that measures the number of people at low, medium, and high seniority levels working in a given firm in a given month.⁹ We then estimate a triple-difference model that interacts our post-ChatGPT dummy, firm-level LLM exposure, and seniority indicators, using the mid-seniority group as our reference group. This specification includes firm-by-period fixed effects, which absorb all time-varying firm-level shocks such as changes in profits, financing costs, industry conditions, management, or any other factor affecting the entire firm that might be correlated with both LLM exposure and employment decisions. We additionally include firm-by-seniority fixed effects (controlling for stable workforce composition preferences) and seniority-by-period fixed effects (capturing aggregate seniority-level trends). The results in Table A10 show strong evidence that employment declines concentrate in low-seniority positions: relative to mid-seniority workers, low-seniority employment at high-exposure firms contracts by an additional 0.7 percent ($p < 0.001$), while high-seniority positions show no significant differential effect. This result confirms the finding that LLM adoption disproportionately affects junior positions (see also Brynjolfsson, Chandar and Chen, 2025; Lichtinger and Hosseini Maasoum, 2025).

⁹Out of 7 seniority classifications in the Revelio Labs data, we classify the lowest two categories as junior (59% of jobs), the next two categories as mid (26%) and the highest three categories as senior (15%).

Third, we test whether our results depend on the minimum firm size threshold used in the sample construction of the firm-level data. Our main analysis restricts to firms with at least 100 employees to ensure consistent LinkedIn coverage and reliable estimation of exposure scores. Tables A11 to A13 relax this restriction to include all firms with at least 10 or 50 employees, expanding the sample to more than 11 million and 2.5 million firm-month observations. The qualitative patterns persist: the stock of low-seniority employment declines, average compensation increases, and exposed firms list significantly fewer jobs in both specifications, although the magnitudes are slightly different.

Fourth, we verify that our results are not sensitive to the task importance scheme used in our main specification. Recall that we follow Eloundou et al. (2024) by assigning weights of 1.0 to core tasks and 0.5 to supplemental tasks when calculating occupation-level exposure scores. Appendix A3.4 replicates our analyses using the continuous task importance scores provided in the O*NET database. The results are quantitatively and qualitatively similar to our main results, suggesting that our findings do not depend on which particular task weighting scheme we use.

Fifth, we estimate weighted regressions, weighing our observations in the firm-level regressions by pre-treatment employment, and in the occupation-level regressions by pre-treatment job posting volume. This approach gives greater weight to occupations representing larger shares of employment to make our estimates more representative for the experiences of the typical worker. Appendix A3.5 presents the results. The findings are largely in line with our main results. For firm-level employment stocks, the weighted specification shows similar overall employment effects, with a slightly larger decline in low-seniority employment (0.6 percent per standard deviation versus 0.4 percent unweighted). For firm-level job postings, the weighted coefficient remains negative but becomes statistically insignificant, suggesting smaller firms drive the hiring freeze response. For occupation-level outcomes, weighted estimates remain nearly identical: job postings decline by 3.8 percent (versus 3.9 percent unweighted) and posted salaries fall by £474 (versus £495 in the main specification). The overall consistency between weighted and unweighted specifications indicates our main conclusions are not driven by extreme outcomes at small entities, with the possible exception of our firm-level job listing result. Finally, we relax the assumption of linear treatment effects by categorizing occupations and firms into exposure quartiles. Tables A20 to A22 present results where the bottom

quartile serves as the reference category. For firm-level job postings, effects are significant across all higher quartiles, though the third and fourth quartiles show similar magnitudes, which is consistent with a threshold effect. For firm-level employment stocks, the evidence is weaker and non-monotonic, with only scattered significant effects across bins. The occupation-level analysis reveals clearer nonlinearities: while the second and third quartiles show minimal effects, the highest exposure quartile experiences significantly larger impacts, suggesting that most of the occupation-level results we observe are driven by the 25 percent most exposed occupations.

5 Conclusion

This paper provides evidence that large language models are reshaping UK labor markets. Using a difference-in-differences design that exploits variation in occupational- and firm-level LLM exposure, we conduct three analyses that show how labor markets change in response to the introduction of generative AI. First, we examine firm-level employment stocks to understand how workforce size and composition evolve within organizations. Second, we analyze firm-level hiring patterns to capture forward-looking recruitment decisions that precede employment adjustments. Third, we investigate occupation-level posting dynamics to identify economy-wide demand shifts.

Our firm-level employment analysis documents a 0.3 percent reduction in total employment per standard deviation increase in LLM exposure, though this effect only becomes statistically significant approximately 18 months after ChatGPT’s introduction. This reduction is predominantly concentrated in junior positions, which decline by 0.4 percent per standard deviation, while senior positions show a smaller, statistically insignificant 0.2 percent reduction. The resulting compositional shift reduces the fraction of junior workers by 0.1 percentage point per standard deviation, mechanically increasing average compensation by £106 as firms retain proportionally more higher-paid senior staff, though this compensation effect is only marginally significant.

While employment stocks adjust gradually, job posting data provide real-time evidence of firms’ evolving labor demand. Firm-level posting activity drops following

ChatGPT’s release, with a one standard deviation increase in LLM exposure reducing the probability of posting any vacancy by 1.0 percentage point. This reduction emerges gradually and becomes significant after approximately 10 months, mirroring the timeline of employment stock adjustments. When we examine the composition of roles driving this reduction, we find that technical positions and creative roles experience the sharpest declines. Sales representative positions, by contrast, show a statistically significant increase, suggesting that customer-facing roles requiring direct interpersonal interaction remain robust to LLM-based automation.

In addition to within-firm dynamics, our occupation-level analysis reveals how labor demand shifts for entire job categories regardless of firm exposure. A one standard deviation increase in LLM exposure is associated with a 3.9 percent reduction in posting volumes, accompanied by £495 (1.1 percent) salary declines. The apparent inconsistency of declining posted salaries for exposed occupations alongside increasing firm-level compensation shows how aggregation could mask opposing forces. While LLM exposure drives down market wages for specific occupations, within-firm rebalancing toward more senior positions pushes average firm-level wages in the opposite direction.

Our heterogeneity analysis reveals that the observed labor market disruptions are almost entirely driven by high-remuneration segments of the labor market. This concentration of effects in higher-wage segments contradicts historical automation patterns that predominantly displaced routine middle-skill work, and suggests that LLMs might lead to a compression of the wage distribution.

Our findings carry important implications for understanding technological displacement and designing policy responses. The concentration of job losses among entry-level positions disrupts traditional skill development pathways where workers master progressively complex tasks through hands-on experience. Without junior roles serving as training grounds, firms may struggle to develop senior talent internally while new entrants face shrinking opportunities to gain experience (Ide, 2025). The freeze in hiring followed by employment stock changes suggests that forward-looking indicators like job postings may provide early warning signals for policymakers. The sharp reductions in technical and analytical roles alongside resilience in sales positions indicates that human interaction and relationship-building skills remain valuable complements to LLM capabilities, suggesting that retraining programs should emphasize these interpersonal competencies. Policy responses

should consider addressing both immediate displacement and longer-term structural changes through, for example, education reform and alternative skill development pathways.

Several important limitations qualify our findings. First, our identification strategy cannot rule out other shocks that differentially affected high- versus low-exposure firms or occupations, such as sector-specific demand shifts, regulatory changes, or complementary technological developments that correlate with LLM susceptibility, although our robustness checks provide evidence against most of these channels. Second, the LinkedIn-based employment data likely overrepresents professional and technical workers who maintain online profiles while underrepresenting manual, service, and informal sector employment, potentially inflating our estimates if LLM impacts concentrate among digitally-active occupations. Third, our exposure measures from Eloundou et al. (2024) capture potential vulnerability to LLM disruption rather than actual adoption, which depends on unobserved factors including financial resources, management quality, technological infrastructure, and organizational culture. Firms with high exposure scores may resist adoption while nominally less-exposed firms might implement LLMs aggressively, attenuating our reduced-form estimates. Finally, our framework cannot observe the creation of entirely new occupational categories enabled by LLM technology such as prompt engineers, AI safety specialists, or human-AI collaboration designers, which historical evidence suggests constitutes a primary channel through which technological change generates employment (Restrepo, 2024).

These limitations highlight the preliminary nature of our findings and provide potential avenues for future research. Most fundamentally, our 30-month observation window may capture only short-term labor market disruption rather than long-term equilibrium outcomes. Previous general-purpose technologies required decades for complementary innovations, business process reengineering, and workforce retraining to fully work their way through the labor market. The initial displacement we document might reverse as organizations develop new LLM-augmented tasks where humans maintain comparative advantage, or intensify as models become more capable and adoption barriers fall. In fact, we already see some evidence that the decrease in hiring in highly exposed occupations attenuates towards the end of the sample period. Future research should extend the analysis period to distinguish between temporary adjustment and permanent structural change. Additionally, future

research could examine heterogeneity in adoption patterns and employment effects across firm characteristics, investigate whether LLM exposure affects job quality and working conditions, and examine a wider set of countries.

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A1 Additional Tables

Table A1: Effect of LLM exposure on firm-level job listings

	Job postings
LLM exposure x Post	−0.010*** (0.002)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. The dependent variable is an indicator equal to one if firm i posted at least one job vacancy in month t . LLM exposure is the firm’s standardized pre-treatment exposure score, calculated as the employment-weighted average of occupational LLM exposure scores from Eloundou et al. (2024) using the workforce composition on 30 October 2022. All regressions include firm fixed effects and industry-by-period fixed effects. All other definitions are as in Table 3.

A2 Heterogeneous Treatment Effects

A2.1 Firm-Level Employment Stocks

Table A2: Effect of LLM exposure on firm-level job outcomes, high vs. low compensation firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: High compensation firms					
LLM exposure x Post	−0.007*** (0.003)	−0.008*** (0.003)	−0.005* (0.003)	−0.001* (0.0004)	195.240* (101.538)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	634,030	633,977	634,018	633,977	634,030
Adj. R-squared	0.975	0.973	0.974	0.963	0.956
Panel B: Low compensation firms					
LLM exposure x Post	0.0001 (0.003)	−0.001 (0.003)	−0.00002 (0.003)	−0.0004 (0.0003)	69.929 (46.938)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	633,983	633,978	632,313	633,978	633,983
Adj. R-squared	0.988	0.988	0.985	0.956	0.946

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for high- and low-compensation firms. Firms are split at the median pre-treatment average compensation level (October 2022). Panel A presents results for firms above the median compensation, Panel B for firms below the median. All other definitions are as in Table 3.

Table A3: Effect of LLM exposure on firm-level job outcomes, London vs. Non-London firms

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
Panel A: HQ in London					
LLM exposure x Post	-0.001 (0.004)	-0.002 (0.004)	-0.004 (0.004)	-0.001 (0.001)	214.807 (132.316)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	377,175	377,129	376,920	377,129	377,175
Adj. R-squared	0.982	0.981	0.981	0.978	0.975
Panel B: HQ outside London					
LLM exposure x Post	-0.004** (0.002)	-0.005** (0.002)	-0.002 (0.003)	-0.001** (0.0003)	71.962 (64.222)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	890,838	890,826	889,411	890,826	890,838
Adj. R-squared	0.983	0.983	0.981	0.979	0.979

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, separately for firms headquartered in London, and those headquartered elsewhere. Panel A presents results for London-based firms, Panel B for firms based outside London. All other definitions are as in Table 3.

A2.2 Firm-Level Job Postings

Table A4: Effect of LLM exposure on firm-level job listings, heterogeneity

	High compensation firms	Low compensation firms	HQ in London	HQ outside London
LLM exposure x Post	-0.013*** (0.003)	-0.007*** (0.002)	-0.016*** (0.003)	-0.008*** (0.002)
Firm fixed-effects	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes
Observations	577,489	690,524	377,175	890,838
Adjusted R ²	0.486	0.554	0.517	0.525

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity across different firm characteristics. Columns 1-2 split firms by median pre-treatment average compensation level. Columns 3-4 split firms by headquarter location (London versus outside London). All other definitions are as in Table A1.

Table A6: Effect of LLM exposure on occupation-level job outcomes, London vs. non-London

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: Job in London				
LLM exposure x Post	-0.034** (0.017)	-562.818*** (139.355)	-0.010*** (0.002)	74.609 (132.525)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,734	16,734	16,734	16,625
Adjusted R ²	0.954	0.922	0.926	0.605
Panel B: Job outside London				
LLM exposure x Post	-0.040** (0.018)	-475.947*** (141.444)	-0.011*** (0.003)	-119.267 (111.131)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,812	16,812	16,812	16,799
Adjusted R ²	0.961	0.938	0.945	0.683

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for London-based jobs and jobs based outside London. Panel A presents results for job openings in London, Panel B for job openings outside London. All other definitions are as in Table 4.

A2.3 Occupation-Level Job Listings

Table A5: Effect of LLM exposure on occupation-level job outcomes, high vs. low salary occupations

	Job postings (log)	Salary	Salary (log)	Salary (SD)
Panel A: High salary occupations				
LLM exposure x Post	-0.057*** (0.022)	-782.371*** (184.426)	-0.013*** (0.003)	-354.844*** (131.069)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,413	8,413	8,413	8,412
Adjusted R ²	0.964	0.903	0.901	0.726
Panel B: Low salary occupations				
LLM exposure x Post	-0.020 (0.031)	-112.614 (179.137)	-0.004 (0.004)	240.391 (170.804)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	8,402	8,402	8,402	8,392
Adjusted R ²	0.961	0.817	0.860	0.436

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, separately for high- and low-salary occupations. Occupations are split at the median pre-treatment average posted salary (October 2022). Panel A presents results for occupations above the median salary, Panel B for occupations below the median. All other definitions are as in Table 4.

A3 Robustness Checks

A3.1 Alternative Fixed Effects

Table A7: Effect of LLM exposure on firm-level job outcomes, time fixed effects

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	-0.00002 (0.001)	-0.004** (0.001)	0.002 (0.002)	-0.002*** (0.0002)	117.995*** (45.546)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.982	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, using year-by-month fixed effects instead of industry-by-period fixed effects. All other definitions are as in Table 3.

Table A8: Effect of LLM exposure on firm-level job listings, time fixed effects

	Job postings
LLM exposure x Post	-0.018*** (0.002)
Firm fixed-effects	Yes
Time fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.519

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity, using year-by-month fixed effects instead of industry-by-period fixed effects. All other definitions are as in Table A1.

Table A9: Effect of LLM exposure on occupation-level job postings, job-role by time fixed effects

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.049*** (0.018)	-556.171*** (141.990)	-0.011*** (0.003)	-105.062 (110.474)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Jobrole-by-period fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.963	0.951	0.955	0.736

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, using job-category (1 out of 7) by year-by-month fixed effects instead of just year-by-month fixed effects. All other definitions are as in Table 4.

A3.2 Within-Firm Analysis of Seniority

Table A10: Effect of LLM exposure on firm-level job outcomes, within firm effects

LLM exposure x Post x High seniority	−0.002 (0.002)
LLM exposure x Post x Low seniority	−0.007*** (0.001)
Firm-by-period FE	Yes
Firm-seniority FE	Yes
Seniority-by-period FE	Yes
Observations	3,792,846
Adjusted R ²	0.992

Notes: This table reports triple-difference estimates of the effect of LLM exposure on employment by seniority level within firms. The dependent variable is log employment at the firm-seniority-month level. We interact the post-ChatGPT indicator and firm-level LLM exposure with seniority indicators, using mid-seniority positions as the reference category. Low seniority corresponds to levels 1–2, mid seniority to levels 3–4, and high seniority to levels 5–7. All regressions include firm-by-period fixed effects, firm-by-seniority fixed effects, and seniority-by-period fixed effects. All other definitions are as in Table 3.

A3.3 Firm Selection

Table A11: Effect of LLM exposure on firm-level job outcomes, min. 10 employees

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	0.0004 (0.001)	−0.002*** (0.001)	0.002*** (0.001)	−0.001*** (0.0002)	160.109*** (52.656)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	11,476,930	11,307,751	10,759,217	11,307,751	11,476,930
Adj. R-squared	0.973	0.967	0.968	0.941	0.944

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes. We use all firms with at least 10 listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table 3.

Table A12: Effect of LLM exposure on firm-level job outcomes, min 50 employees

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure x Post	−0.003** (0.001)	−0.004*** (0.001)	−0.002 (0.002)	−0.0004** (0.0002)	136.552** (53.014)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	2,541,572	2,541,147	2,530,632	2,541,147	2,541,572
Adj. R-squared	0.98	0.978	0.977	0.969	0.966

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes. We use all firms with at least 50 listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table 3.

Table A13: Effect of LLM exposure on firm-level job listings, different minimum firm size thresholds

	min. 10 employees	min. 50 employees
LLM exposure x Post	−0.004*** (0.0003)	−0.008*** (0.001)
Firm fixed-effects	Yes	Yes
Industry-by-period fixed effects	Yes	Yes
Observations	11,476,930	2,541,572
Adjusted R ²	0.487	0.504

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity. We use all firms with at least 10 (column 1) or 50 (column 2) listed employees in October 2022, instead of 100 employees as in our main analysis. All other definitions are as in Table A1.

A3.4 Alternative Task-Importance Weights

Table A14: Effect of LLM exposure on firm-level job outcomes, onet weights

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.002 (0.002)	−0.004* (0.002)	−0.001 (0.002)	−0.001*** (0.0003)	98.318* (56.416)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table 3.

Table A15: Effect of LLM exposure on firm-level job listings, onet weights

LLM exposure x Post	−0.010*** (0.002)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job postings, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table A1.

Table A16: Effect of LLM exposure on occupation-level job outcomes, onet weights

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	−0.040** (0.018)	−352.827*** (128.078)	−0.007*** (0.002)	−31.785 (102.706)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.952	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job postings, using ONET importance weights for tasks instead of the weights used by Eloundou et al. (2024). All other definitions are as in Table 4.

A3.5 Weighted Regressions

Table A17: Effect of LLM exposure on firm-level job outcomes, weighted regressions

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.003 (0.002)	−0.006*** (0.002)	−0.0001 (0.003)	−0.002*** (0.0004)	220.367*** (54.378)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.998	0.998	0.998	0.991	0.991

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, weighting observations by pre-treatment employment (October 2022). All other definitions are as in Table 3.

Table A18: Effect of LLM exposure on firm-level job listings, weighted regressions

LLM exposure x Post	−0.006 (0.008)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adjusted R ²	0.425

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity, weighting observations by the volume of pre-treatment job listings. All other definitions are as in Table A1.

Table A19: Effect of LLM exposure on occupation-level job outcomes, weighted regressions

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	−0.038*** (0.014)	−473.659** (188.293)	−0.011*** (0.003)	−144.807 (169.704)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.952	0.979	0.984	0.871

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job posting, weighting observations by the volume of pre-treatment job listings. All other definitions are as in Table 4.

A3.6 Non-Linear Exposure

Table A20: Effect of LLM exposure on firm-level job outcomes, non-linear exposure

	Employees	Low seniority	High seniority	Low seniority (%)	Av. comp.
LLM exposure bin 2 x Post	0.003 (0.004)	0.002 (0.004)	0.004 (0.004)	−0.001 (0.001)	50.364 (119.736)
LLM exposure bin 3 x Post	−0.003 (0.004)	−0.007* (0.004)	0.003 (0.005)	−0.002*** (0.001)	263.580* (134.525)
LLM exposure bin 4 x Post	0.0004 (0.005)	−0.005 (0.005)	0.006 (0.005)	−0.003*** (0.001)	283.476* (157.186)
Firm fixed-effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes using exposure quartiles. Firms are divided into four bins based on their pre-treatment LLM exposure scores, with bin 1 (lowest exposure quartile) serving as the reference category. Bins 2, 3, and 4 represent the second, third, and fourth quartiles of the exposure distribution, respectively. All other definitions are as in Table 3.

Table A21: Effect of LLM exposure on firm-level job listings, non-linear exposure

LLM exposure bin 2 x Post	−0.015*** (0.004)
LLM exposure bin 3 x Post	−0.025*** (0.005)
LLM exposure bin 4 x Post	−0.026*** (0.005)
Firm fixed-effects	Yes
Industry-by-period fixed effects	Yes
Observations	1,268,013
Adj. R-squared	0.522

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level job posting activity, using exposure quartiles. Firms are divided into four bins based on their pre-treatment LLM exposure scores, with bin 1 (lowest exposure quartile) serving as the reference category. Bins 2, 3, and 4 represent the second, third, and fourth quartiles of the exposure distribution, respectively. All other definitions are as in Table A1.

Table A22: Effect of LLM exposure on occupation-level job listings, non-linear exposure

	Job posts (log)	Salary (lvl)	Salary (log)	SD salary
LLM exposure bin 2 x Post	−0.006 (0.048)	−436.848 (413.685)	−0.004 (0.008)	−248.627 (406.298)
LLM exposure bin 3 x Post	0.033 (0.048)	−298.922 (435.849)	−0.003 (0.008)	3.184 (389.215)
LLM exposure bin 4 x Post	−0.091* (0.047)	−1,018.329** (424.148)	−0.019** (0.008)	−71.245 (392.062)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adj. R-squared	0.962	0.948	0.952	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level job outcomes, using exposure quartiles. Firms are divided into four bins based on their pre-treatment LLM exposure scores, with bin 1 (lowest exposure quartile) serving as the reference category. Bins 2, 3, and 4 represent the second, third, and fourth quartiles of the exposure distribution, respectively. All other definitions are as in Table 4.

A4 Representativeness of Revelio Labs Data

A potential concern with using Revelio Labs data is that the coverage of LinkedIn profiles may not be representative of the broader UK labor market. To investigate this concern, we compare Revelio Labs employment and job posting data against official statistics from the Office for National Statistics (ONS). We conduct two sets of analyses: an aggregate time-series comparison examining whether employment and vacancy trends in Revelio Labs data track official statistics over time, and a cross-sectional industry-level comparison examining whether employment shares across industries are similar in both data sources.

For the aggregate time-series analysis, we standardize both Revelio Labs and ONS monthly time series by subtracting the mean and dividing by the standard deviation, creating z-scores that allow direct comparison of trends. We then regress the standardized ONS employment and vacancy series on their Revelio Labs counterparts. Table A24 presents the results. Column 1 shows that standardized Revelio Labs employment is a strong predictor of standardized ONS employment. Column 2 shows that standardized Revelio Labs job postings similarly track ONS vacancies. The correlations indicate that aggregate employment and hiring dynamics in Revelio Labs data closely mirror official statistics over time.

For the cross-sectional industry-level analysis, we map Revelio Labs' RICS industry classifications to ONS Standard Industrial Classification (SIC) categories. We aggregate employment across 50 Revelio Labs industry categories into 16 broad ONS industry groups, then calculate each industry's share of total employment in both datasets. Table A26 presents regression results from regressing Revelio Labs employment shares on ONS employment shares. The coefficient is again very high, with 0.706. Figure A1 visualizes this relationship: points cluster tightly around the 45-degree line, confirming that industries with larger employment shares in ONS data also have correspondingly larger shares in Revelio Labs data. Taken together, these validation exercises suggest that Revelio Labs data provide a representative picture of UK labor market dynamics.

Table A23: Comparison of ONS and Revelio Labs Employment over Time

Table A24: Comparison of ONS and Revelio employment

	<i>Dependent variable:</i>	
	Employment	Vacancies
	(1)	(2)
Revelio employment	0.456*** (0.121)	
Revelio job listings		0.422*** (0.123)
Constant	0.000 (0.120)	0.184 (0.122)
Observations	56	47
R ²	0.208	0.207
Adjusted R ²	0.193	0.190
Residual Std. Error	0.898 (df = 54)	0.833 (df = 45)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01	

Notes: This table reports OLS estimates from regressing standardized ONS statistics on standardized Revelio Labs data. Column 1 uses monthly employment totals; column 2 uses monthly vacancy/job posting counts. All variables are standardized by subtracting the mean and dividing by the standard deviation.

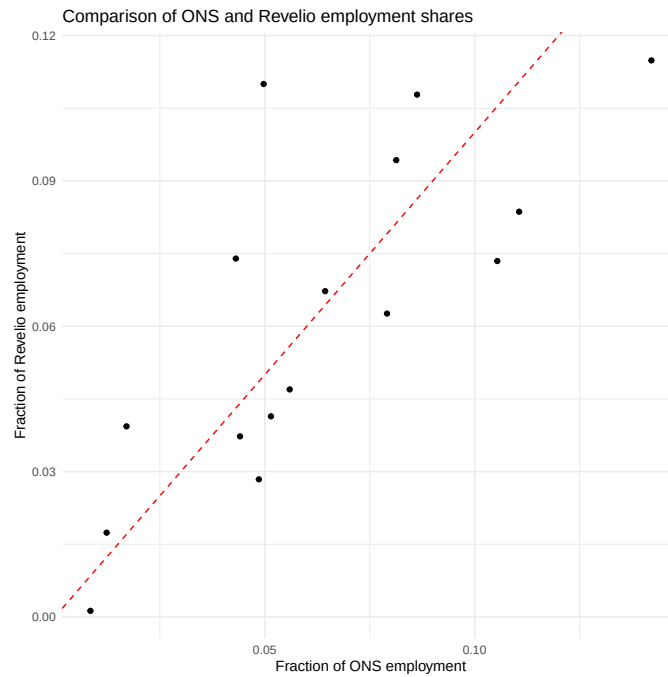
Table A25: Comparison of ONS and Revelio Labs Employment Shares across Industries

Table A26: Comparison of ONS and Revelio employment shares

	<i>Dependent variable:</i>
	Revelio employment
ONS employment	0.706*** (0.161)
ONS job listings	0.018 (0.012)
Observations	16
R ²	0.578
Adjusted R ²	0.548
Residual Std. Error	0.023 (df = 14)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Notes: This table reports OLS estimates from regressing Revelio Labs industry employment shares on ONS industry employment shares. Employment shares are calculated as each industry's fraction of total employment. Industries are defined using 16 broad ONS Standard Industrial Classification (SIC) categories.

Figure A1: Employment Shares: ONS vs. Revelio Labs



Notes: This figure plots industry-level employment shares from Revelio Labs against corresponding shares from ONS data. Each point represents one of 16 broad ONS industry categories. The dashed line indicates perfect correspondence (45-degree line). Points close to this line indicate that an industry's share of total employment is similar in both datasets.